Three ways to run the DSC model

Estimate parameters, smooth states, or hold parameter weights at the prior

The new prior option makes sense as a **prior-sensitivity exercise**: it shows the correlation paths supported by the data when parameter sets retain their prior weights. It answers a different question from posterior estimation.

Mode	Parameters	State paths
Estimation + smoothing	Learn parameters from returns; by default fix them at posterior means for stage 2.	Smooth using all available modeled returns.
Estimation only	Learn and save posterior parameter draws.	Sample states internally; skip the separate smoother and state figures.
Prior + smoothing	Draw parameter sets from the calibrated prior; keep them fixed.	Run separate conditional smoothers for each set; mix them with equal weights.

What the model separates

For weekly returns y_t, the latent states are the changing mean B_t, log variances h_t, and correlation coordinates r_t. The fixed parameters $\theta = (V, \sigma_h^2, \sigma_r^2)$ control how quickly these states move:

$$\begin{aligned} y_t &\sim \mathcal{N}(B_t, D_t P_t D_t), & D_t &= \operatorname{diag}\{e^{h_{jt}/2}\}, \\ B_t - B_{t-1} &\sim \mathcal{N}(0, V), & h_{jt} - h_{j,t-1} &\sim \mathcal{N}(0, \sigma_{h,j}^2), \\ r_{kt} - r_{k,t-1} &\sim \mathcal{N}(0, \sigma_{r,k}^2), & r_t &= \operatorname{vecl}(\log P_t). \end{aligned}$$

P_t is the actual correlation matrix; r_t is an unrestricted numerical coordinate. Smoothing uses future observations relative to a historical week, so old paths may revise when the sample grows.

1. Current default: estimation, then conditional smoothing

1. Reserve the initial 104 weeks for initial-prior calibration and exclude them from the likelihood. Use modeled returns through the estimation cutoff.
2. Run the joint Metropolis-within-Gibbs sampler. A sweep draws B by forward filtering and backward simulation; updates V from its inverse-Wishart conditional; updates each r path by elliptical slice sampling; then proposes σ_r^2 from inverse-gamma distributions with a Metropolis correction for the initial-state density. It updates h and σ_h^2 in the same way.
3. Discard warm-up, retain draws, check diagnostics, and save the parameter snapshot. The default mean-smoothing option takes posterior means of θ.
4. Freeze those means and run state-only chains over the full modeled history, targeting $p(B, h, r \mid y, \widehat{\theta})$. No variance parameters are updated in this stage.

Interpretation. The default stage-2 bands contain conditional state uncertainty; they exclude parameter uncertainty. Each of the two stages has its own `MaxHours` limit. Diagnostics, not the stopping flag, determine whether the saved draws are adequate.

Estimation only, reuse, and calibration

2. Estimate and save parameters only

Set `ParameterSource="posterior"`, `EstimateParameters=true`, and `SmoothStates=false`. Run the same joint sampler described on page 1, check its diagnostics, and save the parameter snapshot. Skip the second conditional smoother and the state figures.

Why are states still sampled? The conditional updates of V, σ_h^2 and σ_r^2 use changes in B, h and r. Estimation only therefore means "no separate smoothing/output stage," not "no latent states inside estimation."

```
result = run_weekly_model_report( ...
  ParameterSource="posterior", EstimateParameters=true, ...
  SmoothStates=false, WarmupIterations=100, ...
  RetainedDraws=100, NumChains=4, ParallelChains=true, ...
  CorrelationBackend="mex_new", MaxHours=12);
```

Examples show option syntax only; no production run was launched for this note.

Reuse saved posterior parameters

Set `EstimateParameters=false`, `ParameterSource="posterior"`, and `SmoothStates=true`. The code loads the newest compatible parameter snapshot and its original prior calibration, then smooths using the available history. Appended observations can change past states without re-estimating the fixed parameters.

The existing `ParameterSmoothing="draws"` alternative. If estimation and smoothing use identical dates, the code can use the joint estimation draws directly: these target the joint posterior once adequately converged. For loaded draws or an extended history, the existing smoother cycles through parameter draws between state sweeps. That is a heuristic; ordinary common-target chain diagnostics are not reported for it. The new prior option on page 3 instead holds each parameter set fixed for an entire conditional chain, with its own warm-up.

Where the prior hyperparameters come from

The priors are **empirical Bayes**: some data calibrate them before the sampler starts. This remains true in the new prior option.

Quantity	Calibration used by the code
$\bar{B}$, $\bar{V}_B$	Fit an intercept-only OLS mean to complete returns in the first 104 weeks. Set $\bar{B}$ to the sample mean and $\bar{V}_B = S_0/n_0$, where S_0 is the covariance with 5% diagonal shrinkage.
$\bar{h}$, $\bar{r}$	From the same S_0: $\bar{h} = \log \operatorname{diag}(S_0)$; standardize S_0 to a correlation matrix and take the lower off-diagonal entries of its matrix logarithm for $\bar{r}$.
V_{0B}, ν_B	$\nu_B = n_0$ complete calibration returns and $V_{0B} = \bar{V}_B \nu_B (0.01)^2$.
a, b_h, b_r	Use 104-week rolling covariance windows through the calibration cutoff. Compute changes in rolling log variances and matrix-log correlations; take their time-series variances, then the median across coordinates within each block. These set blockwise prior means μ_h, μ_r (floored at 10^{-8}). Set $a = 10$ and $b = (a-1)\mu$.

Here n_0 counts complete rows in the reserved window. Initial-state priors are $B_1 \sim \mathcal{N}(\bar{B}, 4\bar{V}_B)$ independently of V and $h_0 \mid \sigma_h^2 \sim \mathcal{N}(\bar{h}, 10\sigma_h^2)$, $r_0 \mid \sigma_r^2 \sim \mathcal{N}(\bar{r}, 10\sigma_r^2)$ componentwise. Integrating the first random-walk step gives $11\sigma^2$ for the first modeled h or r state. The rolling calibration includes later modeled returns through the cutoff, so the exercise is not strictly data-free.

Prior parameter draws + conditional smoothing

3. The new option and its target

Let $X = (B, h, r)$ and let $\pi_{\mathrm{EB}}(\theta)$ denote the calibrated parameter prior. The new option targets

$$\boxed{q(\theta, X \mid y) = \pi_{\mathrm{EB}}(\theta)\, p(X \mid y, \theta).}$$

Returns enter the conditional smoother for X. They do not reweight θ by its likelihood. Consequently q is **not the joint posterior** $p(\theta, X \mid y)$, and it is **not a prior-predictive simulation**, which would generate new data instead of smoothing the observed returns.

1. Draw $K =$ `PriorParameterDraws` independent parameter sets:

$$V^{(k)} \sim \mathrm{IW}(V_{0B}, \nu_B), \quad \sigma_{h,j}^{2(k)} \sim \mathrm{IG}(a, b_h), \quad \sigma_{r,l}^{2(k)} \sim \mathrm{IG}(a, b_r).$$

   The blocks and scalar variance draws are mutually independent.

2. For each k, freeze $\theta^{(k)}$ and start $C =$ `NumChains` conditional state chains. Each chain warms up separately, then retains $R =$ `RetainedDraws` states from $p(X \mid y, \theta^{(k)})$ using the existing B, r and h updates.

3. Check diagnostics *within each fixed parameter set*. Retain the parameter-draw identifiers so conditional convergence and prior mixing remain distinct.

4. With complete equal quotas, estimate a state summary by giving each parameter set weight $1/K$:

$$\widehat{E_q[f(X)]} = \frac{1}{K} \sum_{k=1}^{K} \frac{1}{CR} \sum_{c=1}^{C} \sum_{s=1}^{R} f(X^{(k,c,s)}).$$

Do not redraw θ after one state sweep: a single sweep is generally not a draw from the new conditional smoother.

```
result = run_weekly_model_report( ...
  ParameterSource="prior", PriorParameterDraws=10, ...
  SmoothStates=true, WarmupIterations=100, ...
  RetainedDraws=100, NumChains=4, ParallelChains=true, ...
  CorrelationBackend="mex_new", MaxHours=12);
```

This requests **10 independent parameter sets** and $10 \times 4 \times 100 = 4{,}000$ conditional state draws. All parameter sets share one `MaxHours` stage budget. An incomplete run with unequal component quotas is not pooled as an equally weighted mixture.

How to judge the result

For each fixed $\theta^{(k)}$, the reference checks are $\widehat{R} < 1.01$, bulk ESS and tail ESS ≥ 400, and MCSE/posterior SD ≤ 0.05. Enough retained draws and multiple independent chains are needed for useful checks. Never compute cross-chain $\widehat{R}$ by treating different θ sets as chains of the same target. Good conditional diagnostics do not establish accuracy of the outer prior integral; K controls a separate integration error.

Scope of this version. Prior mode skips posterior parameter estimation and loading, requires `SmoothStates=true`, and does not support automatic extension or resume. The existing publication builder supports posterior runs with smoothing; this standalone note explains all three modes. State figures and saved diagnostic artifacts remain available for prior mode.

Implementation basis: `run_weekly_model_report.m`, `dsc_prepare_inference.m`, `dsc_run_inference.m`, `dsc_sample.m`, and `dsc_calibrate_priors.m` in this branch. No performance, posterior, or convergence results are claimed here.